

Homework (Carolina Reaper)

- Q1.** A music producer uses 2^3 audio tracks. How many tracks will she have if she adds 2^2 more tracks?
(A) 8
(B) 10
(C) 12
(D) 14
(E) 16
- Q2.** A painter's canvas size is 3^2 square feet. If he wants to increase the size by a factor of 3^1 , what will be the new size?
(A) 18
(B) 27
(C) 30
(D) 36
(E) 45
- Q3.** The frequency of a musical note is 2^4 Hz. What is the frequency of the note one octave higher?
(A) 32
(B) 64
(C) 128
(D) 256
(E) 512
- Q4.** Simplify the expression $\frac{a^8}{a^3}$.
(A) a^2
(B) a^5
(C) a^{11}
(D) a^{11}
(E) a^{12}
- Q5.** A rhythm pattern has 3^2 beats. If the pattern is repeated 2^2 times, how many beats will there be in total?
(A) 36
(B) 40
(C) 45
(D) 50
(E) 64
- Q6.** Simplify the expression $(2^2)^3$.
(A) 32
(B) 64
(C) 128
(D) 256
(E) 512
- Q7.** The size of an image is 4^2 pixels. If the image is enlarged by a factor of 2^1 , what will be the new size?
(A) 32
(B) 48
(C) 64
(D) 80
(E) 96
- Q8.** The frequency of a sound wave is 5^2 Hz. What is the frequency of the sound wave one octave higher?
(A) 50
(B) 75
(C) 100
(D) 125
(E) 150

Open Ended Questions (FRQ)

- Q9.** Simplify the expression $\frac{(a^2)^3}{a^4}$. Show all steps and explain your reasoning.
- Q10.** A musical composition has 3^3 notes. If the composer wants to add a 2^3 note melody to the composition, how many notes will the new composition have? Explain your answer and provide the calculation.

Chef's Tips

- Check for accuracy and completeness in calculations
- Ensure students provide clear explanations for their reasoning
- Encourage students to use proper mathematical notation and formatting